Olympiad Test Paper — Science (Class 7) — Paper 4

Chapter 14: Electric Current and Its Effect

Paper 4 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	14 — Electric Current and Its Effect
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Distractors reward shallow rules of thumb — 'more cells always helps', 'a complete-looking loop must light', 'an electromagnet keeps its grip after switch-off', 'a fuse carries excess current away', 'copper heats more than nichrome', 'the bell rings once' — and punish anyone who does not trace the full make-and-break cycle, the branch that current actually takes, or which single effect a device exploits. Do not write on this paper — mark answers on your answer sheet.

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1. A loop is drawn as: battery, then the line splits into two parallel branches that rejoin before returning to the battery. Branch one contains a closed switch and a bulb; branch two contains only a plain wire. With everything connected, the bulb most likely stays dark because:

(A) the plain wire is a low-resistance shortcut beside the bulb, so almost all the current takes that branch and bypasses the filament

(B) the bulb cannot glow whenever a circuit has two branches

(C) the closed switch in the bulb branch cancels the current in both branches

(D) the battery can drive only one branch at a time, and it always chooses the bulb

2. A bulb glows on a battery of two cells joined correctly. A pupil now adds a third identical cell, but inserts it reversed so its push opposes the first two. Compared with the original two-cell brightness, the bulb will most likely:

(A) glow brighter, because any third cell adds to the total push

(B) glow dimmer than with two cells, because the reversed cell subtracts its push from the other two

(C) glow exactly as with two cells, because a reversed cell does nothing at all

(D) go out completely, because one reversed cell cancels two correct cells

3. Two cells of clearly different sizes — a small one and a large one — are joined in a row positive-to-negative the correct way, then put in a loop with a bulb. A pupil insists the bulb cannot glow because the cells are not identical. The pupil is:

(A) right, because only identical cells can ever be joined into a battery

(B) right, because mismatched cells always push against each other

(C) wrong; cells joined the right way add their pushes whether or not they are the same size, so the bulb glows

(D) wrong, but only because the larger cell switches the smaller one off

4. An electromagnet is built on a soft-iron core; an identical coil and current are also wound on a hard-steel core of the same shape. While the current flows, both are roughly equal magnets. After the current is switched off, the difference is that:

(A) both cores instantly lose every trace of magnetism

(B) both cores keep their full strength forever

(C) the soft-iron core stays magnetic while the steel core loses everything

(D) the soft-iron core loses almost all its magnetism, while the steel core keeps a noticeable amount

5. A single loop contains a battery, a closed switch, a good bulb, and a short strip of nichrome heating wire all in series. When the switch is closed, the most accurate prediction is that:

(A) the nichrome blocks all current, so the bulb stays dark

(B) the same current passes through both, the nichrome warms up, and the bulb glows — though a little less brightly than without the nichrome

(C) the bulb glows brighter than normal because the nichrome adds push to the circuit

(D) only the nichrome heats and no current at all reaches the bulb

6. Two heating coils are cut from the same nichrome wire and run on the same battery one at a time. Coil A is long and thin; coil B is short and thick. For the same applied push, which coil draws the larger current and so glows hotter?

(A) Coil A, because a longer wire always carries more current

(B) both equally, because they are made of the same material

(C) neither glows, because nichrome cannot carry current

(D) Coil B, because being short and thick gives it lower resistance, so a larger current flows

7. A house has three circuits, each with its own fuse: lighting (safe up to 5 A), kitchen sockets (safe up to 15 A), and an air-conditioner (safe up to 20 A). An electrician must fit the correct fuse to each. The right choice is to fit:

(A) one 20 A fuse on all three, since the biggest rating is always safest

(B) a fuse rated at each circuit's own safe limit — 5 A, 15 A and 20 A respectively

(C) one 5 A fuse on all three, so every circuit is extra protected

(D) no fuse on the lighting circuit, because small currents are harmless

8. A pupil holds a compass exactly above a straight wire and notices the needle swings one way when current flows. She then runs the same current through the same wire but holds the compass exactly below it. The needle now:

(A) swings the same way as before, because the field is identical all around the wire

(B) does not move at all, because the field exists only on one side

(C) swings the other way, because the magnetic field circles the wire and points oppositely above and below it

(D) spins continuously, because the position has been changed

9. A single loop is drawn with a battery, a bulb, and TWO switches — but the two switches are on separate branches that both connect the same two points (so either branch can complete the loop). For the bulb to glow, it is enough that:

(A) both switches are closed at the same time, or no current flows

(B) both switches are open, so the current is free to pass

(C) at least one of the two switches is closed, since either closed branch completes the loop

(D) exactly one is open and one is closed, to balance the loop

10. An electromagnet on a soft-iron core lifts 6 paper clips with one cell and 30 turns of wire. A pupil rewinds it with 60 turns but, to fit them, uses a much thinner wire that lets less current flow. The number of clips lifted may NOT simply double because:

(A) doubling the turns always exactly doubles the lifting power, whatever the wire

(B) thinner wire has no effect on the magnet's strength at all

(C) more turns strengthen it, but the thinner wire reduces the current, which weakens it — the two changes pull in opposite directions

(D) adding turns weakens every electromagnet, so it will lift far fewer clips

11. In an electric bell, suppose the soft-iron core were secretly swapped for a permanent steel magnet that always stays magnetic. Tracing the cycle, the bell would most likely:

(A) fail to ring repeatedly, because the core never loses its pull, so the armature is held over and the make-and-break cycle cannot restart

(B) ring louder and faster than before, because the magnet is always strong

(C) ring exactly as normal, because the core's magnetism does not matter

(D) ring once and then charge the cell back up

12. A loop has a battery, a closed switch, and TWO identical bulbs joined one after another (in series). A pupil unscrews bulb 1 from its holder, breaking that point of the loop. Bulb 2 will:

(A) glow twice as bright, because all the current now flows through it alone

(B) go dark too, because removing bulb 1 leaves a gap that breaks the single loop and stops the current everywhere

(C) glow normally, because each bulb has its own separate path back to the battery

(D) flash once and then settle to half brightness

13. A pupil places, in turn, into the same single-bulb circuit: a steel paper-clip, a graphite pencil core, a wet cotton string, and a dry plastic comb. In how many cases will the bulb glow at least faintly?

(A) only one — the steel clip, since only metals can conduct

(B) three — the steel clip, the graphite core and the wet string all conduct, while the dry plastic comb does not

(C) all four, because anything placed in the gap completes the loop

(D) none, because a single bulb needs more than one cell to light

14. A coiled tungsten bulb filament and a straight copper lead wire carry exactly the same current. The filament glows white-hot while the copper lead barely warms. The deepest reason the filament gets so much hotter is that:

(A) the copper lead carries no current, so only the filament can heat

(B) tungsten is magnetic and copper is not, and magnetism produces the heat

(C) copper has a higher melting point, so it refuses to get hot

(D) the long, thin, coiled tungsten has far higher resistance than the thick copper lead, so much more heat is produced in it for the same current

15. Two appliances on a wall socket draw a combined current well above the wiring's safe limit, yet nothing trips because someone earlier replaced the proper fuse with a thick nail. The hidden danger is that:

(A) the appliances will simply run at half power and cool down

(B) with no thin link to melt, the overloaded wiring keeps heating and could start a fire instead of being cut off

(C) the nail will magnetise and pull the wires safely apart

(D) the current will reverse direction and protect itself

16. An engineer must confirm whether a sealed cable is carrying current, without breaking its insulation. The single most reliable test that uses an effect of current is to:

(A) bring a small compass near the cable and watch for the needle to deflect when the supply is switched on

(B) feel whether the cable has become lighter when the supply is on

(C) check whether the cable has changed colour

(D) listen for the cable to hum like a bell whenever current flows

17. A coil of 40 turns on a soft-iron core lifts 8 pins on one cell. A pupil keeps the same single cell and the same core but spreads the wire out to only 20 turns. The most likely outcome is that the electromagnet now lifts:

(A) fewer pins, because fewer turns make a weaker electromagnet at the same current

(B) more pins, because fewer turns let the current flow faster

(C) the same number, because only the iron core decides the strength

(D) no pins at all, because an electromagnet needs at least 40 turns

18. In an electric bell the cycle is: current flows, the electromagnet pulls the armature so the hammer strikes the gong, and the same motion breaks the contact. A pupil glues the contact closed so it can never break. With the switch pressed, the bell will:

(A) ring continuously and even faster, because the current can never stop

(B) strike the gong once and stay silent, because the contact never breaks, so the electromagnet holds the armature and the cycle cannot repeat

(C) not ring at all, because gluing the contact opens the circuit

(D) ring normally, because the glue does not affect the cycle

19. A pupil lists which effect each device mainly uses: an electric toaster, an electric bell, and a bulb's glowing filament. The correct mapping is:

(A) toaster — magnetic; bell — heating; glowing filament — magnetic

(B) all three — magnetic

(C) toaster — heating; bell — heating; glowing filament — magnetic

(D) toaster — heating; bell — magnetic; glowing filament — heating

20. A circuit shows a battery, a closed switch, and a bulb. The bulb glows. A pupil then connects a second plain wire directly from one battery terminal to the other, alongside the original loop. The most likely result is that:

(A) the bulb glows brighter, because the extra wire brings it more current

(B) nothing changes, because a spare wire across a battery does nothing

(C) the battery recharges itself through the extra wire

(D) the bulb dims or goes out and the battery drains fast, because the bare wire offers an easy shortcut that carries most of the current

21. A diagram shows a battery, then the wire reaches a switch drawn as a hinged arm tilted up away from its contact, then a bulb, then back to the battery. A pupil reads it as 'closed circuit, bulb glowing'. Why is the pupil wrong?

(A) a hinged arm always means the switch is closed, so the bulb is on

(B) the raised arm shows the switch is open, leaving a gap in the loop, so no current flows and the bulb stays dark

(C) the bulb glows because a battery alone can light it without a complete loop

(D) the tilted arm is a fuse that has melted, but the bulb still glows

22. A pupil claims: 'Replacing one cell with two cells the right way always makes any bulb brighter, with no limit.' The most accurate verdict is that the claim is:

(A) completely correct; more cells always mean more light, with no upper limit

(B) completely wrong; adding cells the right way always makes a bulb dimmer

(C) correct only because the cells share the work between them

(D) mostly right about getting brighter, but wrong about 'no limit' — too many cells can drive enough current to fuse the filament, after which it gives no light

23. An electric fuse is meant to be the deliberately weak link. Compared with the household wiring around it, the ideal fuse wire should have:

(A) a higher resistance and a lower melting point than the household wiring, so it heats fastest and melts first under excess current

(B) a lower resistance and a higher melting point, so it never melts before the wiring

(C) exactly the same resistance and melting point as the wiring, so they melt together

(D) no resistance at all, so it can carry any current safely

24. A heater element and a glowing bulb filament both use the heating effect, yet the bulb gives bright light while a low-temperature heater coil only glows dull red. The key difference is that the bulb's filament:

(A) uses the magnetic effect while the heater uses the heating effect

(B) reaches a far higher, white-hot temperature, so it gives off plenty of visible light

(C) is made of copper, which is why it shines brightly

(D) has lower resistance than the heater, so it stays cool yet shines

25. A loop contains a battery, a closed switch, and an electromagnet (coil on a soft-iron nail) holding several iron pins. The switch is then opened. Tracing the consequences in order, the pins fall because:

(A) opening the switch stops the current, so the coil's magnetic field collapses and the soft-iron core loses its magnetism

(B) opening the switch makes the core a stronger magnet that flings the pins away

(C) opening the switch heats the coil, melting the pins off

(D) gravity suddenly increases the moment the switch opens

26. Two nichrome coils, X and Y, are made of the same length of wire, but Y is twice as thick as X. Each is connected in turn to the same battery. For the same applied push, which coil produces more heat each second, and why?

(A) Coil X, because a thinner wire always produces more heat whatever the current

(B) both equal, because they are the same length and material

(C) Coil Y, because being thicker it has lower resistance, so a larger current flows and more heat is produced

(D) neither, because heat depends only on the cell and not on the wire

27. A pupil wires a circuit so that one straight wire passes directly over a compass and a second identical wire passes directly under the same compass, both carrying current in the same direction. What happens to the needle when current flows?

(A) the needle deflects twice as far, because two wires add their effects

(B) the needle spins continuously, because two fields confuse it

(C) the needle does nothing, because two wires always block all magnetism

(D) the two wires push the needle opposite ways and the deflections largely cancel, so the needle barely moves

28. An electromagnet's pull is tested by hanging a chain of iron nails from it. With one cell it holds 3 nails; with two cells correctly in series it holds 7. A pupil concludes that doubling the cells more than doubled the strength. The fairest comment is that:

(A) the result is impossible, because more cells must weaken an electromagnet

(B) the result is impossible, because cell count cannot affect an electromagnet at all

(C) the nails must have stuck together by themselves without any extra strength

(D) adding the second cell correctly raised the current, which strengthened the electromagnet, so holding more nails is expected

29. In an electric bell, the springy metal strip that carries the armature is replaced with a stiff bar that cannot flex. With current switched on, the bell will most likely:

(A) ring continuously as usual, because the strip's springiness is unimportant

(B) give a single strike at most and then stay silent, because the stiff bar cannot pull the armature back to remake the contact

(C) ring louder, because a stiff bar transmits more force to the gong

(D) never strike at all, because a stiff bar opens the circuit permanently

30. A pupil winds a coil on a soft-iron core but, by mistake, leaves the two ends of the wire NOT connected to any cell. She then holds the coil near iron pins. The pins are:

(A) strongly attracted, because coiling the wire makes it magnetic by itself

(B) not attracted, because with no current there is no magnetic field and the soft iron has no magnetism of its own

(C) attracted only if the core is painted black

(D) attracted, because soft iron is always a permanent magnet

31. A series loop carries a battery, a bulb and a fuse. The fuse is rated just above the normal current. A pupil mistakenly adds extra cells until the current rises sharply. The most likely sequence of events is:

(A) the bulb glows ever brighter and the fuse strengthens to match

(B) the fuse cools down and lets the high current pass safely

(C) the current exceeds the fuse rating, the fuse melts and breaks the loop, and the bulb goes out

(D) the bulb dims while the fuse glows steadily without melting

32. A device is needed that becomes a strong magnet the instant a button is pressed and loses essentially all its magnetism the instant the button is released, over and over. The best choice is:

(A) a permanent steel bar magnet held near the button

(B) a coil with no core at all, relying on the button alone

(C) a tungsten filament that heats when the button is pressed

(D) a coil wound on a soft-iron core connected through the button (an electromagnet)

33. A circuit is drawn as a single loop: battery → bulb 1 → bulb 2 → back to battery, both bulbs glowing. A pupil short-circuits bulb 1 by joining a thick bare wire across its two terminals. The most likely result is that:

(A) both bulbs go dark, because shorting one bulb breaks the whole loop

(B) both bulbs glow brighter, because the extra wire adds current to each

(C) bulb 2 goes dark while bulb 1 glows brighter

(D) bulb 1 goes dark while bulb 2 brightens, because the bare wire bypasses bulb 1 and lets more current flow through bulb 2

34. A pupil compares a fuse and an MCB after a fault has cut off the supply. The clearest practical difference is that:

(A) a fuse can be reset by a switch, while an MCB must always be replaced

(B) both must be thrown away after a single fault

(C) a blown fuse must be replaced with a new one, while a tripped MCB can simply be switched back on once the fault is cleared

(D) neither can ever break a circuit, so both are only decorative

35. Two identical electromagnets are wound with the same wire and the same number of turns and run on the same single cell. Magnet P has a soft-iron core; magnet Q has a wooden core. Which is stronger, and why?

(A) Magnet P, because the soft-iron core greatly increases the coil's magnetic effect, while wood does not

(B) Magnet Q, because wood lets the field spread out more freely

(C) they are equal, because the core never affects an electromagnet

(D) neither, because a coil on any core cannot be magnetic

36. A heating coil glows in a room heater. A pupil claims the coil also produces a magnetic effect while current flows, even though the heater is not meant to be a magnet. This claim is:

(A) correct; any current-carrying wire produces a magnetic effect as well as a heating effect, whether or not the device is built to use it

(B) wrong; a wire can produce either heat or magnetism, never both at once

(C) wrong; only specially shaped coils on iron cores have any magnetic effect

(D) correct, but the magnetic effect appears only after the current is switched off

37. A torch holds two cells. By mistake both cells are inserted reversed, so each faces the opposite way from the correct arrangement — but they still face the same way as each other. Pressing the switch, the bulb will:

(A) not glow, because reversed cells always cancel each other out
(B) glow only half as bright, because reversal halves the push
(C) glow normally, because two cells both reversed still add their pushes in one direction, just the opposite way round the loop
(D) glow brighter than usual, because reversal doubles the push

38. A pupil draws an electric-bell cycle as a list and asks which single arrow is wrong: current flows → electromagnet pulls armature → hammer strikes gong → contact breaks → current stops → electromagnet keeps its full magnetism → armature held forever. The wrong step is:

(A) 'current flows' — current never starts a bell cycle
(B) 'electromagnet keeps its full magnetism' — once the current stops, the soft-iron core loses its magnetism and releases the armature
(C) 'hammer strikes gong' — the hammer is not part of the cycle
(D) 'contact breaks' — the contact stays closed throughout

39. An electromagnet is used at a recycling plant to pull one metal out of a moving mixture. Which mixture can it usefully sort by attracting one part and leaving the rest?

(A) copper coins mixed with brass washers, by attracting the copper
(B) glass chips mixed with plastic flakes, by attracting the glass
(C) steel bottle-caps mixed with aluminium cans, by attracting the steel caps and leaving the aluminium
(D) aluminium foil mixed with paper, by attracting the aluminium

40. A loop contains a battery, a closed switch, and a bulb, and the bulb glows steadily. A pupil now opens the switch for a moment and closes it again. During the brief moment the switch is open, the bulb:

(A) goes dark, because the open switch breaks the loop so no current flows, then relights the instant the switch closes again
(B) stays glowing, because the bulb stores enough current to keep shining
(C) glows brighter, because opening the switch builds up extra current
(D) is permanently destroyed by the brief interruption

41. A pupil compares the filament of a bulb with the connecting wires leading to it. Both carry the same current, yet only the filament glows. The best single explanation is that the filament:

(A) is a thin, high-resistance tungsten wire, so for the same current it heats to a far higher temperature than the thick, low-resistance leads
(B) carries far more current than the leads, which is why it heats up
(C) is magnetic, and the magnetism makes it glow
(D) is made of a better conductor than the leads, so it heats more

42. A pupil wires a battery to a coil-and-soft-iron electromagnet through a bell-style make-and-break contact, but forgets to fit any hammer or gong. With the switch pressed, what will happen?

(A) nothing will move at all, because a bell needs a gong to start the cycle

(B) the armature will be pulled across once and then stay locked there

(C) the armature will still vibrate back and forth as the contact makes and breaks, though no sound is produced without a gong

(D) the coil will overheat and melt because there is no gong to absorb the energy

43. A pupil claims: 'Because a fuse melts to protect a circuit, fitting a fuse with a much HIGHER rating gives even better protection.' This claim is:

(A) correct; the higher the rating, the safer the circuit always is

(B) correct, because a high rating makes the fuse melt sooner

(C) wrong only because high-rated fuses cost more money

(D) wrong; a higher-rated fuse only melts at a much larger current, so it lets a dangerous overload pass before acting — it protects worse, not better

44. Two compasses are placed near the same straight current-carrying wire, one to the left of it and one to the right, both at the same height as the wire. When current flows, the two needles:

(A) deflect in exactly the same direction, because the field is the same everywhere

(B) both point straight at the wire, because the wire attracts them

(C) do not move, because side-by-side compasses cancel the field

(D) deflect in opposite senses, because the circular field around the wire points oppositely on its two sides

45. A pupil joins three cells in a row but, distracted, makes the order: forward, forward, reversed, all in series with a bulb. Compared with all three forward, the bulb will:

(A) glow brighter, because three cells are present whatever their direction

(B) glow the same, because the number of cells is unchanged

(C) glow less brightly, because the reversed cell subtracts its push, leaving a net of about one cell

(D) not glow at all, because any reversed cell stops the current completely

46. A pupil tests three loops, each with a battery and a bulb. Loop 1 also has dry wood in series; loop 2 also has a wet rag in series; loop 3 also has a copper strip in series. In which loops does the bulb glow at least faintly?

(A) only loop 3, because nothing but metal can ever conduct

(B) all three, because any object placed in a loop completes it

(C) loops 2 and 3, because the wet rag and the copper both conduct, while dry wood blocks loop 1

(D) none, because adding anything in series always stops a bulb

47. A pupil wants the brightest possible glow without melting the bulb and lists four changes. The single change most likely to brighten the bulb safely is to:

(A) replace the connecting copper wire with a length of dry string

(B) add one more cell correctly in series, raising the push and so the current a little

(C) add a long thin nichrome coil in series with the bulb

(D) join a bare wire straight across the bulb's terminals

48. A circuit is wired so that current must flow through an electromagnet's coil and then through a separate bulb, all in one series loop. When a fault increases the current sharply, both the bulb's brightness and the electromagnet's pull are seen to rise together. The best explanation is that:

(A) the current splits, sending more to the bulb and less to the coil

(B) the bulb and coil each generate their own current independently

(C) the same larger current passes through both, heating the filament more (brighter bulb) and strengthening the coil's field (stronger pull)

(D) the coil steals the current from the bulb, so only one can strengthen

49. A pupil holds a compass under a wire and sees the needle deflect by a small angle when current flows. He then joins a second cell correctly in series, raising the current, with the wire and compass unmoved. The needle now:

(A) deflects through a smaller angle, because more current weakens the field

(B) deflects through a larger angle, because a larger current makes a stronger magnetic field around the wire

(C) deflects by the same angle, because deflection does not depend on the current

(D) stops deflecting, because two cells cancel the magnetic effect

50. A pupil claims an immersion water heater and a ceiling fan both depend on the same effect of current. The correct response is that:

(A) they are the same — both use the heating effect

(B) they are the same — both use the magnetic effect

(C) they differ — the heater uses the magnetic effect and the fan uses the heating effect

(D) they differ — the immersion heater works on the heating effect, while the fan's motor works on the magnetic effect

51. A loop has a battery, a closed switch, and a bulb that glows. A pupil predicts that adding a second closed switch one-after-another (in series) in the same loop will brighten the bulb. The pupil is:

(A) wrong; an extra closed switch in series merely keeps the loop complete and does not add push, so the brightness is essentially unchanged

(B) right; every extra closed switch adds push and brightens the bulb

(C) right; a second switch doubles the current

(D) wrong; a second switch always darkens the bulb completely

52. A pupil tests a fuse wire on its own across a strong battery and watches it glow, then melt and break. Ranking the moments, the correct order of what happens is:

(A) the wire melts first → then heats → then current begins to flow

(B) current flows → the thin wire heats up → it glows → it reaches its melting point and melts → the circuit breaks and current stops

(C) the wire glows → cools down → carries the current forever

(D) the circuit breaks first → then current flows → then the wire melts

53. A current-carrying coil on a soft-iron core lifts iron filings. A pupil reverses the connections to the cell so the current flows the opposite way through the same coil. The electromagnet will:

(A) stop being a magnet, because reversing the current cancels the field

(B) become much stronger, because reversed current adds to the field

(C) still attract the iron filings just as strongly, because reversing the current keeps the magnet's strength the same even though its poles swap

(D) turn into a heater instead of a magnet

54. In an electric bell, a pupil files down the contact screw so the gap at the contact is far too wide ever to touch, even when the armature is fully at rest. With the switch pressed, the bell will:

(A) ring continuously and faster, because the wide gap speeds up the cycle

(B) ring exactly once, because the electromagnet always fires a single strike

(C) ring louder, because a wide gap stores up extra current

(D) not ring at all, because the over-wide gap means the contact never closes, so no current ever reaches the electromagnet

55. A pupil lists statements and is asked which one is INCORRECT about effects of current. Which statement is wrong?

(A) A current can produce a magnetic effect without ever producing any heating effect in an ordinary resistive wire.

(B) A current in any resistive wire produces some heating effect, however small.

(C) A current in a wire produces a magnetic effect that a compass can detect.

(D) Increasing the current generally increases both the heat in a wire and the field around it.

56. A pupil builds two electromagnets to lift more iron. Magnet A: 50 turns, one cell, soft-iron core. Magnet B: 50 turns, two cells correctly in series, soft-iron core. Predict which lifts more and why.

(A) Magnet A, because a single cell concentrates its push into fewer turns

(B) they lift equally, because both have 50 turns and a soft-iron core

(C) Magnet B, but only because two cells add extra turns to the coil

(D) Magnet B, because the two cells drive a larger current through the same coil and core, making a stronger electromagnet

57. A pupil draws a circuit symbol as a small circle with a cross inside, placed in a closed loop with a battery and a closed switch, but the battery's two cells inside it are drawn facing each other (positive-to-positive). What does the circled cross stand for, and what should it do?

(A) it is a bulb, but it stays dark because the two opposing cells cancel each other's push, so almost no current flows

(B) it is a fuse, and it melts at once because the opposing cells make a huge current

(C) it is a bulb, and it glows normally because the loop is closed

(D) it is a switch, so the loop is broken and nothing can work

58. A pupil compares the heat made each second in a thin nichrome coil for three settings of the same circuit: one cell, two cells correctly in series, three cells correctly in series. As cells are added the right way, the heat produced each second:

(A) increases at each step, because more cells drive a larger current and a larger current makes more heat

(B) decreases at each step, because the cells share the heating between them

(C) stays the same, because heat depends only on the nichrome and not on the cells

(D) first rises and then falls back to the one-cell value

59. A pupil wires a bell so the make-and-break contact is in series with the coil, exactly as normal, but accidentally connects the cell with very weak, nearly dead cells. With the switch pressed, the most likely behaviour is that the bell:

(A) rings louder than normal, because weak cells make the cycle faster

(B) rings perfectly normally, because cell strength does not affect a bell

(C) rings weakly or not at all, because the feeble current makes too weak an electromagnet to pull the armature firmly

(D) melts the coil, because weak cells force a larger current

60. A pupil claims: 'Since both a fuse and an electric bell contain a wire that current flows through, both protect a circuit by breaking when the current is too high.' This claim is:

(A) wrong; a fuse breaks the circuit by melting under excess current, but a bell uses the magnetic effect to ring and is not a protective device

(B) correct; both melt to break the circuit when current is too high

(C) wrong; neither device ever breaks a circuit under any condition

(D) correct; a bell protects a circuit while a fuse only makes sound

Solutions — Science (Class 7), Chapter 14: Electric Current and Its Effect — Paper 4

Paper 4 — (progressively harder)

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	A	21	B	31	C	41	A	51	A
2	B	12	B	22	D	32	D	42	C	52	B
3	C	13	B	23	A	33	D	43	D	53	C
4	D	14	D	24	B	34	C	44	D	54	D
5	B	15	B	25	A	35	A	45	C	55	A
6	D	16	A	26	C	36	A	46	C	56	D
7	B	17	A	27	D	37	C	47	B	57	A
8	C	18	B	28	D	38	B	48	C	58	A
9	C	19	D	29	B	39	C	49	B	59	C
10	C	20	D	30	B	40	A	50	D	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (A) The two branches join the same two points, so they sit side by side across the battery. Current prefers the easy path, so the bare wire branch drains nearly all of it and the filament gets too little to glow. Branching itself does not forbid glowing, a closed switch does not cancel current, and a battery does not 'choose' a branch — it feeds both, just unequally.

2. (B) Pushes add when cells face the same way and subtract when one faces the other way. Two forward and one reversed leaves a net push of about one cell's worth, weaker than the two-cell total, so the bulb dims but does not die. It cannot brighten, a reversed cell is not inert, and one cell cannot fully cancel two.

3. (C) Joining positive-to-negative makes the pushes add; matching sizes is not required for a working battery. The cells do not oppose each other when wired the same way, and neither switches the other off. So the bulb lights.

4. (D) Soft iron magnetises and demagnetises easily, so it drops its pull when current stops; steel holds on to some magnetism, which is why permanent magnets are made of steel-like metals. So it is the steel, not the soft iron, that retains magnetism after switch-off.

5. (B) In one series loop the same current flows through every part. Nichrome conducts (it does not block current) but its resistance lowers the current a little, so the bulb glows slightly dimmer rather than brighter, and the nichrome heats up. A resistive element adds no push and does not starve the bulb of all current.

6. (D) Resistance rises with length and falls with thickness. The short, thick coil has the lower resistance, so the larger current flows through it and it gets hotter. Same material does not mean same resistance, and nichrome certainly carries current.

7. (B) A fuse must match the safe current of the circuit it guards: too high and it never melts to protect that circuit, too low and it blows during normal use. So each circuit needs its own rating. A single 20 A fuse leaves the lighting circuit unprotected, a single 5 A fuse blows on the heavier circuits, and even small-current circuits need a fuse.

8. (C) The magnetic field forms circles around the wire, so its direction is opposite above and below the conductor; the needle therefore deflects the other way. The field is not one-sided, nor identical all round, and a steady current does not make the needle spin.

9. (C) Because the two switches sit on parallel branches between the same points, closing either one bridges the gap and completes the loop, so one is enough. This differs from switches placed one-after-another, where every one must be closed. Both open leaves no path, and 'balancing' is not how switches work.

10. (C) Strength grows with both the number of turns and the current. Doubling the turns helps, but the thinner wire carries less current, which hurts, so the result is a tug-of-war and not a clean doubling. Wire thickness affects current and hence strength, and adding turns alone does not weaken a magnet.

11. (A) Repeated ringing needs the core to lose its magnetism each time the contact breaks, so the spring can pull the armature back and remake the contact. A permanently magnetic core keeps gripping the armature, freezing the cycle. A stuck armature does not ring louder, the core's switching is essential, and bells do not recharge cells.

12. (B) In a single series loop there is only one path; removing one bulb opens that path, so no current flows and the surviving bulb also goes dark. The bulbs do not have separate paths, and an open loop carries no current to make bulb 2 brighter or flash.

13. (B) Steel (a metal), graphite (a conducting non-metal), and damp string all let some current through; dry plastic is an insulator. So three of the four light the bulb. Conduction is not limited to metals, dry plastic blocks the loop, and one cell can light a bulb.

14. (D) For the same current, the part with higher resistance releases more heat; the slender, coiled, high-resistance tungsten therefore reaches white heat while the low-resistance copper stays cool. Both carry the same current in a series path, heat comes from resistance and not magnetism, and copper's melting point does not stop it warming.

15. (B) A fuse protects by melting and breaking an overloaded circuit; a thick nail never melts, so nothing stops the dangerous current and the wiring overheats toward a fire. Overload does not reduce appliance power, a nail does not pull wires apart, and current does not reverse to self-protect.

16. (A) A current always sets up a magnetic field, so a nearby compass needle deflects when current flows — a clean, non-invasive check (the magnetic effect). Current does not change a cable's weight or colour, and an ordinary cable does not buzz like a bell.

17. (A) More turns mean a stronger electromagnet for a given current, so halving the turns weakens it and it lifts fewer pins. Fewer turns do not speed up current usefully, the turn-count does matter alongside the core, and there is no fixed 40-turn threshold.

18. (B) Repeated ringing depends on the contact breaking to release the armature; with it glued shut, the electromagnet stays on, holds the armature, and the hammer rests against (or beside) the gong after one strike. A permanently closed contact does not open the circuit, and it certainly does affect the cycle.

19. (D) The toaster and the glowing filament both rely on current heating a high-resistance element, while the bell rings using an electromagnet — the magnetic effect. The mappings that call the bell a heater or the filament a magnet are mixed up.

20. (D) A bare wire straight across the terminals is a near short circuit: it has very low resistance, so it grabs most of the current and the bulb is starved while the battery runs down quickly. The shortcut does not feed the bulb more, it does change things, and batteries do not recharge through a shorting wire.

21. (B) A switch drawn with its arm lifted off the contact is open, so the loop is broken and the bulb cannot light. A raised arm is not 'closed', a battery cannot light a bulb across an open gap, and the symbol is a switch, not a melted fuse.

22. (D) Adding cells correctly does raise the push and the current, so the bulb brightens — up to a point. Past that point the current is so large the filament melts and the bulb stops glowing, so 'no limit' is false. Correctly added cells do not dim a bulb, and they do not 'share' the work to weaken it.

23. (A) To melt first and protect the rest, the fuse must heat up faster (higher resistance) and give way sooner (lower melting point) than the wiring it guards. A tougher fuse than the wiring, or an identical one, would not melt in time, and a zero-resistance wire would never heat at all.

24. (B) The hotter an object glows, the whiter and brighter its light; the filament reaches white heat, so it lights brightly, while a cooler element only glows dull red. Both use the heating effect, the filament is tungsten not copper, and a brightly glowing filament is very hot, not cool.

25. (A) No current means no magnetic field, and soft iron keeps almost no magnetism on its own, so the pins are released. Opening a switch removes the field rather than strengthening it, it stops the heating rather than melting pins, and gravity is unchanged.

26. (C) At a fixed push, the thicker, lower-resistance coil lets a larger current through, and the larger current makes more heat per second. Thinness alone does not win when the current it allows is much smaller, equal length does not force equal heat, and the wire's own properties clearly matter.

27. (D) A wire's field points one way above it and the opposite way below it, so a wire above and a wire below the compass push the needle in opposite directions and the effects nearly cancel. They do not add to double the swing, a steady current does not spin the needle, and current still produces magnetism.

28. (D) More cells the right way means more current, and more current makes a stronger electromagnet, so lifting more nails is exactly what should happen. More cells do not weaken it, cell count does affect it, and the extra grip comes from real added strength, not chance sticking.

29. (B) The spring's job is to pull the armature back after the contact breaks so the contact remakes and the cycle repeats; a rigid bar cannot do this, so the cycle stalls. The springiness is essential, stiffness does not make it ring louder repeatedly, and a stiff bar does not by itself open the circuit forever.

30. (B) An electromagnet needs current to flow; without a connected cell there is no current, no field, and soft iron alone is not magnetic, so nothing is picked up. Coiling alone does not magnetise, paint is irrelevant, and soft iron is not a permanent magnet.

31. (C) Too many cells drive a current above the fuse's rating; the fuse then melts and opens the loop, so the bulb loses its current and goes dark. A fuse does not strengthen, it does not cool to pass excess current, and its role is to melt — not glow steadily — under overload.

32. (D) An electromagnet with a soft-iron core turns strongly magnetic with current and demagnetises almost completely without it, perfect for switching on and off. A steel magnet cannot be switched off, a coreless coil is far weaker, and a heated filament is about light and heat, not magnetism.

33. (D) The bare wire gives current an easy path around bulb 1, so bulb 1 is starved and dark; with less total resistance in the loop, more current flows through bulb 2 and it brightens. Shorting one bulb does not open the loop, and it is bulb 1 — the bypassed one — that loses its glow, not bulb 2.

34. (C) A fuse works by melting and so is single-use, but an MCB trips a switch that can be reset after the fault is fixed. So the fuse is the one that needs replacing, not the MCB; both do break overloaded circuits and are far from decorative.

35. (A) A soft-iron core concentrates and multiplies the coil's magnetism, but a wooden core adds nothing, so P is far stronger. Wood does not enhance the field, the core does matter, and a current-carrying coil is certainly magnetic.

36. (A) Every current sets up a magnetic field and also heats a resistive wire, so both effects are present together — the heater simply does not use its (weak) magnetic effect. A current does not choose one effect, the magnetic effect is not limited to iron-cored coils, and it is present while the current flows, not after.

37. (C) If both cells face the same way, their pushes add regardless of which way that is; the current simply circulates the other way round and the bulb glows as normal. Cells cancel only when they face each other, and reversing both does not halve or double the push.

38. (B) When the contact breaks the current stops, so the core demagnetises and the spring returns the armature — the magnetism cannot persist. Current does start the cycle, the hammer striking the gong is the whole point, and the breaking contact is exactly what restarts the cycle.

39. (C) Electromagnets attract iron and steel but not copper, brass, glass, plastic, paper or aluminium, so only the steel-and-aluminium mixture can be sorted by pulling out the steel. The other mixtures contain no magnetic metal for the electromagnet to grab.

40. (A) An open switch breaks the only path, so current stops at once and the bulb goes dark, returning to normal the moment the loop is closed again. A bulb does not store current to keep glowing, opening a switch does not build up current, and a brief break does not destroy the bulb.

41. (A) In a series path the same current flows through filament and leads alike; the filament's high resistance makes it far hotter, so it glows while the low-resistance leads stay cool. The filament does not carry extra current, magnetism is not the cause, and it is the higher resistance — not better conduction — that produces the heat.

42. (C) The make-and-break cycle drives the armature to vibrate whether or not a gong is present; you simply hear no ring. The gong is not what starts the cycle, the armature does not lock (the contact still breaks), and a missing gong does not cause the coil to melt.

43. (D) A fuse should melt at the circuit's safe limit; raising the rating lets a far larger, unsafe current flow before the fuse responds, so protection gets worse. A higher rating does not make it melt sooner, and the real fault is safety, not cost.

44. (D) The wire's field circles around it, so on opposite sides the field direction is opposite and the two needles swing opposite ways. The field is not uniform all round, the wire does not simply attract the needles, and the compasses do not cancel each other's view of the field.

45. (C) Forward pushes add and the reversed one subtracts, so two forward minus one reversed leaves roughly one cell's push — dimmer than three forward. The mere count of cells does not fix brightness, direction matters, and one reversed cell reduces rather than fully stops the current here.

46. (C) Copper conducts well and a wet rag conducts enough for a faint glow, but dry wood is an insulator and blocks its loop. So two of the three light. Conduction is not limited to metals, an insulator does break a loop, and a conductor in series does not stop the bulb.

47. (B) One more cell correctly in series increases the push and brightens the bulb without the huge jump that would fuse it. Dry string is an insulator and kills the glow, a nichrome coil in series adds resistance and dims the bulb, and a bare wire across the bulb shorts it out and darkens it.

48. (C) In a single series loop one and the same current flows through every part, so a larger current both brightens the filament (heating effect) and strengthens the electromagnet (magnetic effect) at once. The current does not split in a single loop, the parts do not make their own current, and neither steals current from the other.

49. (B) A larger current produces a stronger field, so the needle swings further. More current strengthens rather than weakens the field, the deflection does depend on the current, and two cells in series add to the current rather than cancelling it.

50. (D) An immersion heater warms water through a high-resistance element (heating effect), while a fan spins because of an electric motor that uses the magnetic effect. So the two devices use different effects, and the heater is the heater while the fan is the magnetic one.

51. (A) A switch only makes or breaks a loop; a second closed switch in series simply leaves the loop complete, adding no push, so brightness stays about the same. Switches do not add push or double current, and a closed switch does not darken the bulb.

52. (B) Heating must precede melting: the current heats the thin, high-resistance wire until it glows, then melts, which finally breaks the loop and stops the current. Melting cannot precede heating, a melting wire does not cool to carry current forever, and the break is the result of melting, not its cause.

53. (C) Reversing the current swaps which end is north and which is south but does not change how strongly the electromagnet pulls iron, so it lifts the filings just the same. Reversal does not cancel or boost the field, nor convert the magnet into a heater.

54. (D) If the contact can never close, the circuit never completes, so the electromagnet never energises and the bell stays silent. A permanently open contact does not speed up or store current, and with no current the electromagnet cannot even fire once.

55. (A) Any current through an ordinary resistive wire heats it at least a little, so a current cannot give a magnetic effect with literally no heating — that statement is the false one. The others are true: resistive wires do warm up, a compass detects the field, and more current means more heat and a stronger field.

56. (D) With the same turns and core, the magnet carrying the larger current is stronger, and two cells in series give more current, so B lifts more. A single cell does not concentrate push, equal turns alone do not equalise strength when the currents differ, and cells add current — not turns — to the coil.

57. (A) The circled cross is the bulb symbol, but opposing cells cancel their pushes so the loop carries hardly any current and the bulb stays dark. A closed loop alone does not guarantee a glow when the cells oppose, opposing cells give little current rather than a huge one, and the circled cross is not a switch.

58. (A) More cells the right way mean a larger current, and a larger current makes more heat per second, so the heating rises with each added cell. Cells do not 'share' to reduce heat, the driving current does matter, and there is no reason for the heat to fall back.

59. (C) The electromagnet's pull depends on the current; nearly dead cells give too little current to work the armature, so the bell rings feebly or stays silent. Weak cells do not speed up or normalise the cycle, and they give less current — not more — so the coil does not melt.

60. (A) A fuse is a safety device that melts to break an overloaded circuit, while a bell rings using an electromagnet and is not built to protect anything — so the claim conflates two unrelated jobs. A bell does not melt for safety, the fuse certainly does break circuits, and the bell-versus-fuse roles are not reversed.